of swaps (the proof is easy enough if you count what's the worst-case possible number of swaps in a path where all elements are the same – details are left to the reader).

2.6.7 Heapify

Priority queues can be created empty, but they are also often initialized with a set of elements. If that's the case, if we need to initialize the heap with n elements, we can still obviously create an empty heap and add those elements one by one.

To do so, we will need at most $O(n)$ time to allocate the heap array, and then repeat n insertions. Every insertion is logarithmic, so we have an upper bound of $O(n \log n)$.

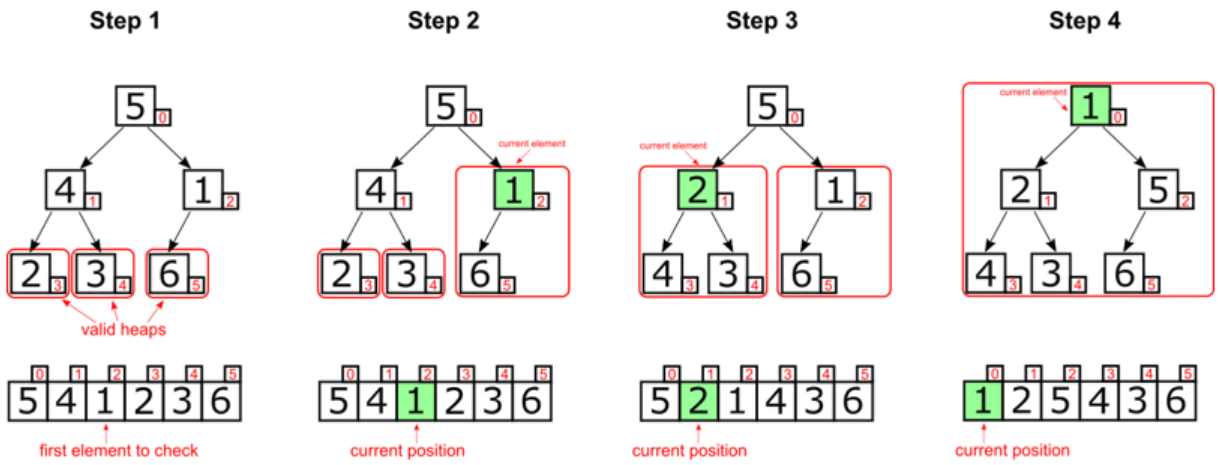


Figure 2.12 Heapification of a small array. The red boxes surround valid sub-heaps, i.e. portions of the tree for which we have verified that they abide by the heap properties.
(Step 1) Initially, only leaves are valid heap.
(Step 2) The element from which we start our